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The Fractional Fourier Transform on Graphs: Sampling and Recovery

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***Abstract*—Signal processing on graphs expands discrete signal processing theory and techniques to signals supported on graphs. In this paper, we study the sampling and recovery of graph signals under the graph fractional Fourier transform. We show that a-bandlimited signals in the graph fractional Fourier domain can be perfectly recovered. Experimentally designed sampling strategy is used to generate optimal fractional sampling operators on graphs. We give numerical examples, and test the semi-supervised classification of online blogs and handwritten digits using fractional sampling on graphs, and compare it with GFT sampling. We find that fractional sampling on graphs can lead to better classification accuracy at an optimal fractional order.**

***Keywords*—Signal processing on graphs, the fractional Fourier transform, sampling theory, semi-supervised learning**

I. Introduction

Big data processing often deals with high volume of data residing on complex network structures. These datasets, unlike time series or digital images, possess complex and irregular structures, which makes utilizing conventional signal processing tools rather difficult. The demand for novel techniques eventually leads to the emergence of signal processing on graphs [1], [2].

Signal processing on graphs, also known as graph signal processing, generalizes conventional discrete signal processing to signals with complex, irregular underlying structures characterized by corresponding graphs. Unlike other fields of study in complex network analysis, signal processing on graphs emphasizes the interaction between signals and their graph structures, and aims to build up a theoretical framework for real-world data analysis task from a signal processing perspective. The framework extends concepts and tools in discrete signal processing, including transforms on graphs [3], [4], [5], frequency analysis [6], sampling and interpolation [7], [8], graph-based filtering [9], fast computation algorithms [10], and so on.

There are two fundamental approaches to signal processing on graphs: The first one emerges from *spectral graph theory* [1], and mainly bases on *graph Laplacian matrix*. This approach, however, is confined to undirected graphs with real non-negative weighted edges, because the standard graph Laplacian matrix must be symmetric and positive semi-definite. The second one is *discrete signal processing on graphs* ($\mathrm{DSP_G}$) [2], [3], which derives from *algebraic signal processing theory* and bases on the graph shift operator. The graph shift operator, i.e. the weighted adjacency matrix, works as the fundamental building block of any linear shift-invariant filter for signals on a given graph [7]. The adjacency matrix is not restricted to be symmetric, hence this approach can be applied to arbitrary graphs, whose edges can either be directed or undirected, with negative or non-negative, real or complex weights. In this paper, we follow the $\mathrm{DSP_G}$ framework for its broader scope.

An important tool in $\mathrm{DSP_G}$ is the graph Fourier transform, which is defined as expanding the graph signal onto a basis of eigenvectors of the adjacency matrix. It is the direct generalization of the discrete Fourier transform onto graphs. In particular, the graph Fourier transform reduces to discrete Fourier transform for signals on cycle graphs [3]. Various processes can be performed on real-world graph signals whose energy is more concentrated to low frequencies in the graph Fourier domain.

Besides Fourier transform in classical signal processing, the fractional Fourier transform was proposed in 1980 [11], and introduced to the signal processing community later on in 1994 [12]. It is another generalization of the Fourier transform, which rotates a signal to an intermediate domain between time and frequency. The fractional Fourier transform can also be viewed as expanding a signal onto a set of chirp signals. Its application includes filtering, radar and optical signal processing.

In our previous study [13], we extended the fractional Fourier transform onto graphs and proposed the graph fractional Fourier transform (GFRFT). We proved that it satisfies all the expected properties and calculated two examples of GFRFT. In this paper, we further consider the sampling and interpolation task within GFRFT. The classical Shannon sampling theory connects continuous functions to discrete series, and shows that a bandlimited function can be perfectly recovered from its sampled series when the sampling rate is high enough. In the broad sense, any linear operator resulting in a decrease in dimension can be viewed as sampling, and that resulting in an increase in dimension as interpolation. Sampling theory for discrete signals in this context can be interpreted as moving between higher- and lower-dimensions [7].

In [14], the authors demonstrated that bandlimited graph signals can be perfectly recovered with sufficient sample size. Further, three types of sampling strategies for graph signals in previous works, including uniform sampling, experimentally designed sampling [7], and active sampling[8], [15], were compared in terms of the minimax lower bounds on recovery error. Their results showed that active sampling achieves the

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same rate of convergence as experimentally designed sampling strategy, while outperforming uniform sampling for approximately bandlimited graph signals on irregular graphs.

In this paper, we propose and consolidate the fractional sampling theory on graph following [7] and [13]. Section II briefly introduces the $\mathrm{DSP_G}$ framework and our previous work in [13]. Section III extends the sampling theory in [7] to graph fractional domain. Section IV discusses the choice of optimal sampling operator, and gives numerical examples of fractional sampling on graphs to validate the theory. Section V tests the performance of classification of online blogs and handwritten digits using fractional sampling on graphs, and compares it with GFT sampling. Finally, section VI concludes the paper.

II. The Fractional Fourier Transform on Graphs

In this section, we briefly introduce concepts in the discrete signal processing on graphs framework [2], [3] and our previous work on the graph fractional Fourier transform [13].

A. Discrete Signal Processing on Graphs

In discrete signal processing on graphs, the high-dimensional structure of a signal is represented by a graph $\mathcal{G} = (\mathcal{V}, \mathrm{A})$, where $\mathcal{V} = v_0, \ldots, v_{N-1}$ denotes the set of vertices and $\mathrm{A} \in \mathbb{C}^{N\times N}$ denotes the weighted adjacency matrix. The edge $\mathrm{A}_{m,n}$ weighs the directed relation from vertex v_m to v_n. When the graph is undirected, in other words the relation is bidirectional, we have $\mathrm{A}_{n,m} = \mathrm{A}_{m,n}$, which makes A symmetric. To scale the signal properly, the adjacency matrix is normalized to satisfy $|\lambda_{\max}(\mathrm{A})| = 1$.

Given this graph, the *graph signal* is defined as a mapping that maps the vertex v_n to a signal coefficient $x_n \in \mathbb{C}$, which can also be written as a complex vector

$$x = \begin{bmatrix} x_0 & x_1 & \ldots & x_{N-1} \end{bmatrix}^T \in \mathbb{C}^N.$$

While the classical Fourier transform expands a signal onto a basis of exponent signals, *the graph Fourier transform* expands it on to the eigenvectors of the adjacency matrix A (or the Jordan eigenvectors if the complete eigenbasis of A does not exist). For simplicity, we assume A has a complete eigenbasis and has the following spectral decomposition

$$\mathrm{A} = \mathrm{V}\mathrm{J}\mathrm{V}^{-1}, \tag{1}$$

where the columns of V are the eigenvectors of A, corresponding to eigenvalues $\lambda_0, \ldots, \lambda_{N-1}$ in the diagonal matrix $\mathrm{J} \in \mathbb{C}^{N\times N}$. The eigenvalues are the *graph frequencies* in descending order from low to high frequency [6].

Definition 1. The *graph Fourier transform* (GFT) of $s \in \mathbb{C}^N$ is defined as

$$\hat{s} = \mathrm{V}^{-1}s, \tag{2}$$

and the *inverse graph Fourier transform* (IGFT) is

$$s = \mathrm{V}\hat{s}, \tag{3}$$

where V^{-1} is the GFT matrix, and vector $\hat{s}$ represents the signal's frequency coefficients in the graph Fourier domain.

B. The Fractional Fourier Transform

The The ath-order fractional Fourier transform (FRFT) [11], [12] can be interpreted as a linear operator that rotates the time axis to frequency axis by an angle of $\phi = a\pi/2$ (rather than $\pi/2$ as in Fourier transform). There are three categories of approach to define the discrete fractional Fourier transform (DFRFT), including linear combination [16], sampling [17], and eigenvalue decomposition [18], [19]. In [20], the three categories are evaluated in terms of the following properties:

1) Unitarity;
2) Index additivity;
3) Reduction to Fourier transform when $a = 1$;
4) Resemblance to the continuous FRFT.

The authors showed that all three categories of DFRFT reduce to Fourier transform when $a = 1$, and their other properties are listed in Table 1.

Using eigenvalue decomposition method, the DFRFT matrix is given by

$$\mathrm{F}^a[m,n] = \sum_{k=0}^{N-1} u_k[m]\mathrm{e}^{j\frac{\pi}{2}ka}u_k[n], \tag{4}$$

with $u_k[n]$ denoting the kth discrete Hermite-Gaussian series, which is the eigenfunction of the Fourier transform [18]. The $\mathrm{e}^{j\frac{\pi}{2}ka}$ term can be interpreted as the eigenvalue of the discrete Fourier transform matrix ($\mathrm{e}^{j\frac{\pi}{2}k}$) to the power of a.

TABLE I
Comparing three categories of DFRFT

Properties	Linear Combination	Sampling	Eigenvalue Decomp.
Unitary	√	√	√
Index Additive	√	×	√
Resemble FRFT	×	√	√

C. Graph Fractional Fourier Transform

Parallel to the eigenvalue decomposition method of defining discrete fractional Fourier transform, we proposed in [13] the definition of graph fractional Fourier transform.

Definition 2. The ath-order *graph fractional Fourier transform* (GFRFT) of $x \in \mathbb{C}^N$ is defined as

$$\hat{x} = \mathrm{F}^a = \mathrm{Q}\Lambda^a\mathrm{Q}^{-1}x, \quad a \in [0,1], \tag{5}$$

and the inverse GFRFT is

$$x = \mathrm{F}^{-a} = \mathrm{Q}\Lambda^{-a}\mathrm{Q}^{-1}\hat{x}, \tag{6}$$

where matrix Q and $\Lambda \in \mathbb{C}^{N\times N}$ are given by the spectral decomposition of GFT matrix

$$\mathrm{V}^{-1} = \mathrm{Q}\Lambda\mathrm{Q}^{-1}. \tag{7}$$

The term Λ^a is computed by applying the power a to every entry in the diagonal matrix Λ. It is easy to verify that when $a = 0$, GFRFT of a graph signal is the signal itself, and when $a = 1$, GFRFT reduces to GFT. Other properties of GFRFT can be found in [13].

TABLE II
KEY NOTATIONS IN THE PAPER

Symbol	Description	Dimension
A	adjacency matrix	$N \times N$
x	graph signal	N
V^{-1}	graph Fourier transform matrix	$N \times N$
a	fractional order	1
F^a	graph fractional Fourier transform matrix	$N \times N$
$\hat{x}$	transformed graph signal	N
Ψ	sampling operator	$M \times N$
Φ	interpolation operator	$N \times M$
$\mathcal{M}$	sampled indices	
$x_{\mathcal{M}}$	sampled signal	M
$\hat{x}_{(K)}$	fist K coefficients of $\hat{x}$	K
$\mathrm{F}_{(K)}^{-a}$	fist K columns of F^{-a}	$N \times K$

III. Fractional Sampling on Graphs

In this section, we propose the fractional sampling theory on graphs following the GFT sampling theory in [7].

A. Sampling & Interpolation

Consider sampling M coefficients from a graph signal $x \in \mathbb{C}^N$ to obtain a sampled signal $x_{\mathcal{M}} \in \mathbb{C}^M (M < N)$, where $\mathcal{M} = \{\mathcal{M}_0, \ldots, \mathcal{M}_{N-1}\}, \mathcal{M}_i \in \{0, 1, \ldots, N-1\}$ denotes the set of sampled indices. The sampling operator Ψ is defined as a linear mapping from $\mathbb{C}^N$ to $\mathbb{C}^M$,

$$\Psi_{i,j} = \begin{cases} 1, & j = \mathcal{M}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

We then recover x from $x_{\mathcal{M}}$ with interpolation operator Φ, which is a linear mapping from $\mathbb{C}^M$ to $\mathbb{C}^N$.

$$\begin{aligned} \text{sampling}: &\quad x_{\mathcal{M}} = \Psi x \in \mathbb{C}^M, \\ \text{interpolation}: &\quad x' = \Phi x_{\mathcal{M}} = \Phi \Psi x \in \mathbb{C}^N, \end{aligned}$$

where $x' \in \mathbb{C}^N$ recovers x either approximately or exactly.

B. Fractional Sampling Theory for Graph Signals

We now define a class of a-bandlimited signals under the graph fractional Fourier transform.

Definition 3. A graph signal is *a-bandlimited* when there exist a $K \in 1, \ldots, N$ such that the transformed signal $\hat{x}$ satisfies

$$\hat{x}_k = 0 \quad \text{for all} \quad k \geq K - 1.$$

We call the smallest such K the *a-bandwidth* of x.

Definition 4. The set of graph signals in $\mathbb{C}^N$ with a-bandwidth no greater than K is denoted by $\mathrm{BL}_K(\mathrm{F}^a)$, with F^a as in (5).

We have the following theorem parallel to theorem 1 in [7].

Theorem 1. For Ψ such that

$$\mathrm{ramk}(\Psi \mathrm{F}_{(K)}^{-a}) = K, \quad K < N \tag{9}$$

where $\mathrm{F}_{(K)}^{-a}$ denotes the first K columns of F^{-a}. For all $x \in \mathrm{BL}_K(\mathrm{F}^a)$, perfect recovery, $x = \Phi\Psi x$, is achieved by choosing

$$\Phi = \mathrm{F}_{(K)}^{-a} \mathrm{U},$$

where $\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a} = \mathrm{I}_{K \times K}$ is a $K \times K$ identity matrix.

Proof. Since $\mathrm{rank}(\Psi\mathrm{F}_{(K)}^{-a}) = K$ and $\mathrm{rank}(\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a}) = K$, $\mathrm{rank}(\mathrm{U}) = K$, i.e. U spans $\mathbb{C}^K$. Hence, $\Phi = \mathrm{F}_{(K)}^{-a}\mathrm{U}$ spans $\mathrm{BL}_K(\mathrm{F}^a)$.

Let $\mathrm{P} = \Phi\Psi$, we have

$$\begin{aligned} \mathrm{P}^2 = \Phi\Psi\Phi\Psi &= \mathrm{F}_{(K)}^{-a} \left(\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a}\right) \mathrm{U}\Psi \\ &= \left(\mathrm{F}_{(K)}^{-a} \mathrm{I}_{K\times K} \mathrm{U}\right) \Psi = \Phi\Psi = \mathrm{P}, \end{aligned}$$

so $\mathrm{P} = \Phi\Psi$ is a projection operator.

Since $\Phi\Psi$ is a projection and Φ spans $\mathrm{BL}_K(\mathrm{F}^a)$, $\Phi\Psi x$ is an approximation of x in $\mathrm{BL}_K(\mathrm{F}^a)$. When x itself is in $\mathrm{BL}_K(\mathrm{F}^a)$, $\Phi\Psi x = x$, i.e. perfect recovery is achieved. □

There is a lower bound for the sampling size $M \geq K$, because when $M < K$, $\mathrm{rank}(\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a}) \leq \mathrm{rank}(\mathrm{U}) \leq M < K$, so $\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a}$ can never be an identity matrix. For $\mathrm{U}\Psi\mathrm{F}_{(K)}^{-a}$ to be identity matrix, U should be the inverse of $\Psi\mathrm{F}_{(K)}^{-a}$ when $M = K$, and the pseudo-inverse of $\Psi\mathrm{F}_{(K)}^{-a}$ when $M > K$. For simplicity, we only consider $M = K$ and invertible U.

We call the sampling operator Ψ that satisfies (9) a *qualified* sampling operator. A qualified sampling operator should select at least K linearly-independent rows in $\mathrm{F}_{(K)}^{-a}$. The first K columns of F^{-a} must be linearly-independent, since F^{-a} is invertible and thus full-ranked. So there is always at least one combination of K linearly-independent rows in $\mathrm{F}_{(K)}^{-a}$. Hence, theorem 1 guarantees perfect recovery for a-bandlimited graph signals.

IV. The Choice of Sampling Operator

The previous section shows that a qualified sampling operator, which is obtained by selecting K linearly-independent rows in $\mathrm{F}_{(K)}^{-a}$, can guarantee a perfect recovery for a-bandlimited graph signals. The choice of such rows is not unique, and we use in this section the experimentally designed sampling strategy to achieve maximum robustness to noise.

A. Optimal Sampling Operator

As there are multiple choices of K linearly-independent rows in $\mathrm{F}_{(K)}^{-a}$, we aim to choose the optimal set that minimizes the influence of noise. Consider noise e introduced when sampling

$$x_{\mathcal{M}} = \Psi x + e,$$

with Ψ being qualified, the recovered signal, x'_e, would then be

$$x'_e = \Phi x_{\mathcal{M}} = \Phi\Psi x + \Phi e = x + \Phi e.$$

So the error of recovery is bounded by

$$\begin{aligned} ||x' - x||_2 &= ||\Phi e||_2 = ||\mathrm{F}_{(K)}^{-a}\mathrm{U}e||_2 \\ &\leq ||\mathrm{F}_{(K)}^{-a}||_2 ||\mathrm{U}||_2 ||e||_2. \end{aligned}$$

Since $||\mathrm{F}_{(K)}^{-a}||_2$ and $||e||_2$ are fixed, we aim to minimize the spectral norm of U, which is the pseudo inverse of $\Psi\mathrm{F}_{(K)}^{-a}$. In other words, we maximize the smallest singular value $\sigma_{\min}$ of $\Psi\mathrm{F}_{(K)}^{-a}$ for every qualified Ψ,

$$\Psi^{\mathrm{opt}} = \arg\max_{\Psi} \sigma_{\min}(\Psi\mathrm{F}_{(K)}^{-a}). \tag{10}$$

We call the solution of (10) *optimal sampling operator* for it achieves maximum robustness to noise. The optimization problem can be solved using a greedy algorithm [7] as shown in algorithm 1.

Algorithm 1 Optimal Sampling Operator via Greedy Algorithm

Input $\mathrm{F}_{(K)}^{-a}$ the first K columns of $\mathrm{F}_{(K)}^{-a}$
M the number of samples
N the number of rows in $\mathrm{F}_{(K)}^{-a}$

Output $\mathcal{M}$ sampling set

Function

for $i = 1 : M$
$m = \arg\max_{j \in \{1:N\} - \mathcal{M}} \sigma_{\min}\left((\mathrm{F}_{(K)}^{-a})_{\mathcal{M}+\{j\}}\right)$
$\mathcal{M} = \mathcal{M} + \{m\}$
end
return $\mathcal{M}$

B. Simulations

We simulate a 64 nodes David sensor network using the Graph Signal Processing ToolBox (GSPBox) [21] in MATLAB. We consider the real part of a graph signal with 0.7-bandwidth of 3 as

$$x = \mathrm{real}(\mathbf{f}_1 + 0.5\mathbf{f}_2 + \mathbf{f}_3)$$

where $\mathbf{f}_i$ denotes the ith column in F^{-a}. Details of the original signal are given in Figure 1.

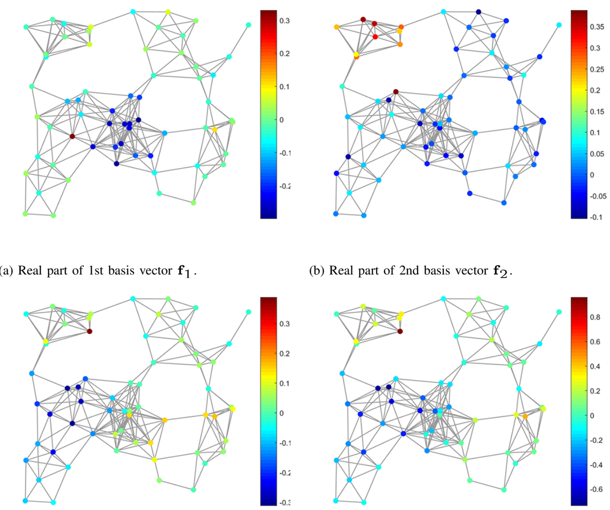
We sample and interpolate the graph signal with two optimal sampling operators based on 0.7th-order GFRFT and GFT respectively, and then measure the accuracy of recovery with $\varepsilon = ||x' - x||_2$. Figure 2 shows the sampled coefficients. Figure 3 (a) shows the perfectly recovered signal through fractional sampling ($\varepsilon = 3.6796 \times 10^{-16}$), while (b) shows the recovered signal from GFT sampling ($\varepsilon = 1.3846$). The GFT sampling fails to recover the graph signal perfectly, because the signal is, though 0.7-bandlimited, not strictly bandlimited in the graph Fourier domain.

V. Applications

Semi-supervised classification aims to classify a large amount of data from a small number of labels. From the graph signal processing perspective, the labels are assume to be smooth on the graph, and smoothness is reflected by being low-pass in graph frequency domain. In this scenario, the task is equivalent to recovering low-pass graph signal. In this section, we test the classification of online blogs and handwritten digits with fractional sampling and recovery, and compare it with GFT sampling.

A. Classifying Online Blogs

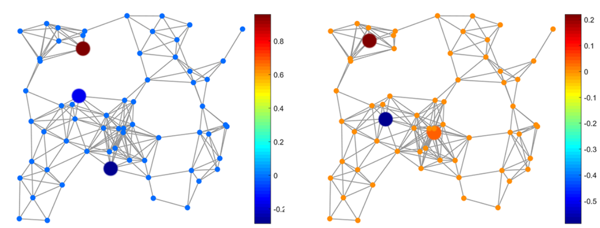
We try to classify the political blog dataset, which consists of $N = 1224$ online blogs as either conservative or liberal [23]. The nodes in the graph represent the blogs and the edges represent the hyperlinks between blogs. Here, we assign the labelling signal to conservatives as 0 and liberals as 1. The labelling signal is full-band, but approximately a-bandlimited when $a \approx 1$. Hence, we approximate the labelling signal with



(a) Real part of 1st basis vector $\mathbf{f}_1$. (b) Real part of 2nd basis vector $\mathbf{f}_2$.

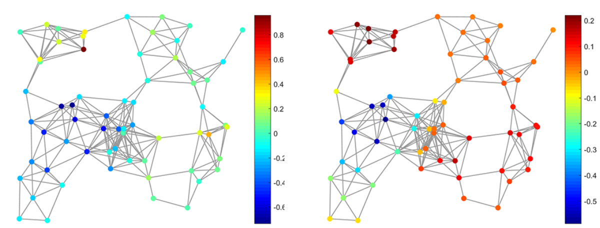
(c) Real part of 3rd basis vector $\mathbf{f}_3$. (d) Original graph signal.

Fig. 1. Graph signals on the David sensor network.



(a) Samples from fractional sampling. (b) Samples from GFT sampling.

Fig. 2. Sampled signals. Big nodes represent the sampled ones.



(a) Recovery from fractional sampling. (b) Recovery from GFT sampling.

Fig. 3. Recovered signals on the David sensor network.

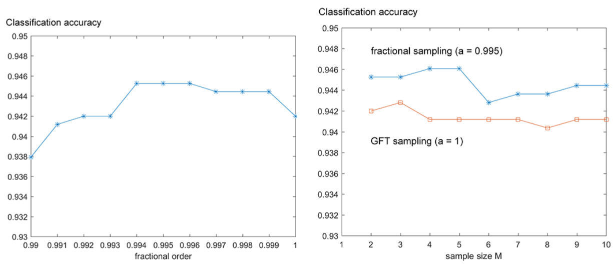
the K lowest fractional frequency components only by solving the following optimization problem,

$$\hat{x}_{(K)}^{opt} = \arg \min_{\hat{x}_{(K)} \in \mathbb{R}^K} \left\| \mathrm{thres}\left(\Psi \mathrm{F}_{(K)}^{-a} \hat{x}_{(K)}\right) - x_{\mathcal{M}} \right\|_2^2, \quad (11)$$

where the threshold function $\mathrm{thres}(\cdot)$ assigns all values greater than 0.5 to 1, and others to 0. The threshold function can be relaxed with the logistic function, and the problem can be solved with logistic regression [24]. We then set another 0.5 threshold to assign labels from the recovered signal,

$$x^{opt} = \mathrm{thres}\left(\mathrm{F}_{(K)}^{-a} \hat{x}_{(K)}^{opt}\right). \quad (12)$$

Figure 4 (a) shows the variation of classification accuracy by sampling $M = K = 2$ nodes when we vary the fractional order a, notice that $a = 1$ corresponds to GFT sampling. The accuracy reaches a maximum value of 94.5261% when $a = 0.994, 0.995, 0.996$, which is higher than 94.1993% when $a = 1$. (b) shows that 0.995th order fractional sampling outperforms GFT sampling in terms of classification accuracy as M varies.



(a) Classification accuracy as a function of fractional order a.

(b) Classification accuracy as a function of sample size M.

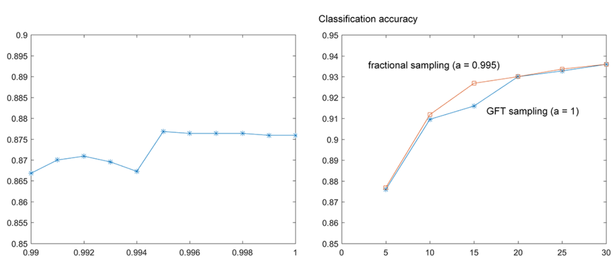
Fig. 4. Classification of online political blogs.

B. Classifying Handwritten Digits

We select the digit "4" and "9" in the USPS handwritten digits dataset [25] for classification, as the two digits can often be mistaken for each other in real life. The dataset includes 1100 sample images with $16 \times 16 = 256$ pixels for each digit. We first construct a 6-nearest neighbor graph to represent the similarity between images, which is measured by the Euclidean distance. The adjacency matrix of the 6-nearest neighbor graph is given by $A = \mathrm{P}_{i,j} / \sum_{i=1}^{6} \mathrm{P}_{i,j}$, where

$$\mathrm{P}_{i,j} = \begin{cases} \exp\left(\frac{-N^2 \|y_i - y_j\|_2}{\sum_{i,j} \|y_i - y_j\|_2}\right), & \text{for 6 largest elements in each row,} \\ 0, & \text{otherwise,} \end{cases}$$

where y_i is a vector that represents the digital image. We label the sampled "4"s with 0 and the "9"s with 1, use logistic regression to solve (11) and assign all labels with (12).



(a) Classification accuracy as a function of fractional order a.

(b) Classification accuracy as a function of sample size M.

Fig. 5. Classification of USPS handwritten digit "4" and "9".

Figure 5 (a) shows the classification accuracy by sampling $M = K = 5$ nodes as the fractional order a varies. The accuracy reaches a maximum value of 87.6818% when $a = 0.995$, which is higher than 87.5909% when $a = 1$. (b) compares the classification accuracy of 0.995th order fractional sampling and GFT sampling as M varies. We can see that 0.995th order fractional sampling outperforms GFT sampling except for $M = 20, 30$ where the two have the same accuracy.

VI. Conclusion

In this paper, we proposed the fractional sampling theory on graphs that generalizes the sampling theory in [7]. We showed that a-bandlimited graph signals in the fractional Fourier domain can also lead to perfect recovery through ath order GFRFT, and the experimentally designed sampling strategy was adopted to guarantee perfect recovery while achieving maximum robustness to noise. We then tested the fractional sampling and recovery using numerical examples, as well as semi-supervised classification datasets. When we compared its performance with GFT sampling, we found that fractional sampling can lead to better classification accuracy at an optimal fractional order.

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